

CIFAKE robustness project, handover

INTERNAL WORKING DOCUMENT, NOT SUBMISSION-READY. For Katie, from Will, 8 August. Deadline is Friday 14 August, 5pm.

The question we are answering

Can a CNN that detects AI-generated images keep working when the images are degraded? Detectors are usually tested on clean images, but real images that need checking have been compressed and resized on their way through the internet. We train two copies of the same CNN on CIFAKE (60,000 real CIFAR-10 images vs 60,000 Stable Diffusion fakes, 32x32 pixels). The baseline copy trains on clean images only. The robust copy trains with corrupted images mixed in (data augmentation). We then test both on progressively degraded test images and see whose accuracy survives.

What changed from our proposal, and why

The proposal said we would augment with blur and noise. The notebook now augments with noise and JPEG compression instead, and keeps blur out of training entirely. The reason is that testing a model on the same corruptions it trained on only shows memorisation. Holding blur out gives us a corruption the robust model has never seen, so any advantage there is real generalisation. This mirrors what Hendrycks and Dietterich (2019) do in the CIFAR-10-C benchmark paper we cite. The marker feedback also asked for metrics beyond accuracy, so every run now records precision, recall, F1 and ROC-AUC as well (all in the CSVs).

The frozen protocol (do not change without re-running everything)

Both strategies share every setting except augmentation. Same CNN (two conv blocks then two fully connected layers, 545k parameters), same 90/10 train/validation split, same Adam optimiser at learning rate 0.001, up to 30 epochs with early stopping (patience 5) against a clean validation set, weights restored from the best epoch. Three seeds (42, 43, 44); each seed gives both models the identical split and identical starting weights, so every comparison is paired. Evaluation covers 16 conditions: clean, plus 5 severities each of Gaussian noise, JPEG compression and Gaussian blur. Blur is evaluation-only. All numbers below are test-set accuracy, mean plus or minus sample standard deviation over the 3 seeds.

Proposed ResNet-18 protocol (not yet implemented, nothing run)

Will's agreed contribution is a second baseline-versus-robust comparison using ResNet-18 (torchvision, the architecture from the week 5 transfer-learning lab), answering the marker's ask for a wider range of architectures. The proposal is to mirror the frozen CNN protocol exactly where it matters for comparability: the same 90/10 train/validation split, the same three seeds (42, 43, 44) with paired baseline/robust runs per seed, the same noise and JPEG training augmentations with blur held out, the same 16-condition evaluation, and the same five metrics per condition. Early stopping stays patience 5 against the clean validation set with best-weights restore. Open choices to settle before running: pretrained ImageNet weights versus training from scratch, how to feed 32x32 images to an ImageNet-shaped network (upsampling versus adapting the first layer), and whether the learning rate needs lowering for fine-tuning. Nothing is coded or trained yet; this section exists so we agree the design before any compute is spent.

Results (CNN runs, complete)

Condition (worst severity shown)	Baseline	Robust	Gap
Clean	0.9442 ± 0.0014	0.9491 ± 0.0012	+0.005
Noise sigma 0.10 (trained-on)	0.8029 ± 0.0030	0.9338 ± 0.0030	+0.131
JPEG quality 40 (trained-on)	0.8826 ± 0.0087	0.9282 ± 0.0038	+0.046
Blur sigma 1.0 (held-out)	0.6382 ± 0.0284	0.7401 ± 0.0061	+0.102

Three things worth saying out loud. First, robustness cost nothing on clean images (the robust model is actually 0.005 better). Second, the headline result is blur. The robust model never saw blur in training, yet it beats the baseline by 10 to 15 accuracy points at blur sigma 0.6 and above. That is generalised robustness, not memorised augmentation, and it is the claim the report should be built around. Third, every gap exceeded the observed between-seed standard deviation. The baseline spread at blur 1.0 is 0.028; the gap there is 0.10. The baselines also stopped early at 9 or 10 epochs on all three seeds while the robust runs went to 17 or 19, consistent with augmentation acting as regularisation (a link to the week 3 overfitting lecture for the report).

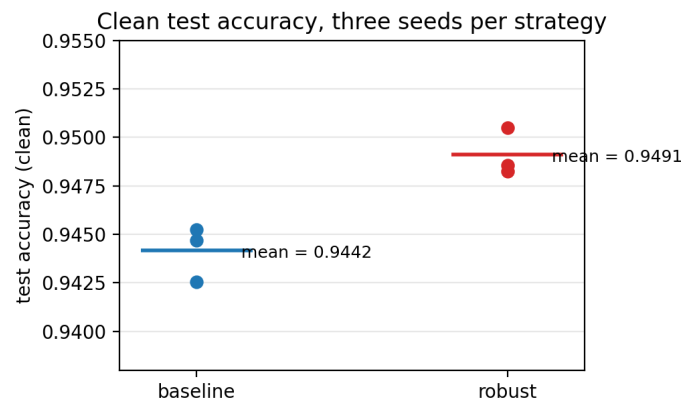


Figure 1. Clean test accuracy per seed. Bars mark 3-seed means.

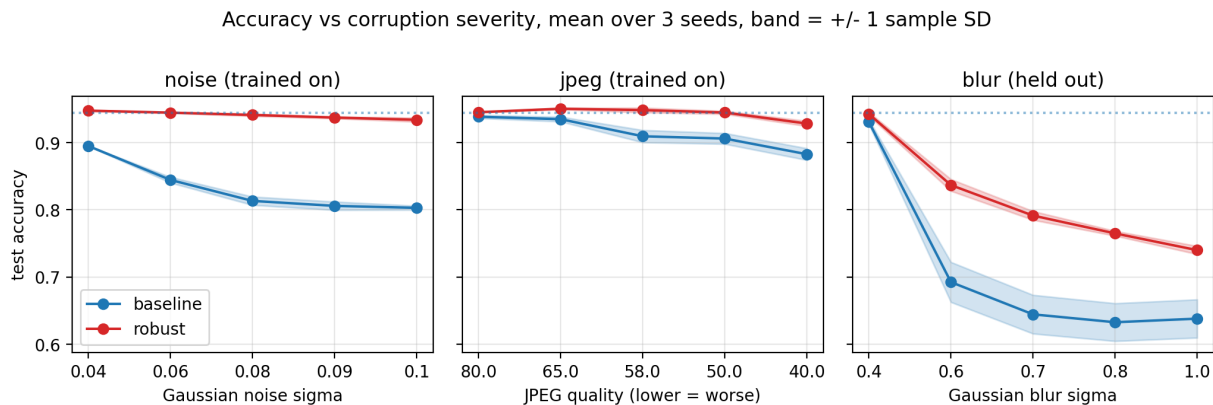


Figure 2. Accuracy against severity for the three corruption types. Shaded band is plus or minus one sample SD over seeds. Dotted line is the baseline clean accuracy. The right panel (blur) is the held-out corruption.

Limitations we must own in the report

- 1) One architecture so far, one dataset, 32x32 images; we cannot claim this transfers to full-resolution detectors. The agreed ResNet-18 comparison addresses the architecture half once run.
- 2) Blur is one held-out family; a stronger design would rotate the held-out corruption.
- 3) The corruption severities are our own picks calibrated by eye against CIFAR-10-C, not a standard benchmark.
- 4) Three seeds is enough to show the gaps exceed the observed between-seed standard deviation but too few for anything fancier.
- 5) CIFAKE fakes come from one generator (Stable Diffusion 1.4), so "fake" here is narrower than in the wild.

Files, all under Assignment/group project/

220231053_251104382_Group_Neural_Networks_Cifake.ipynb is the single notebook (runs top to bottom in about 5 seconds with `run_training = False`; it loads saved results rather than retraining). `results/` holds the six accepted run bundles: for each of baseline/robust x seeds 42/43/44 there is a checkpoint (e.g. `baseline_seed42.pt`), a training history (`baseline_seed42_history.json`) and a 16-condition metric table (`baseline_seed42_results.csv`), plus the combined `all_results.csv` (96 rows, 5 metrics per row). Every bundle passed the notebook's validation gate (config match, dataset version match, condition set, loadable weights). `Group 10 Project Proposal.pdf` and `reply-to-proposal.jpeg` are the proposal and the feedback.

Remaining work

The ResNet-18 comparison is agreed as Will's. Everything else is unassigned; let's split it on our next call.

- [] [Will] ResNet-18 baseline-versus-robust comparison under the proposed protocol above (implement, run all three seeds, validate bundles).
- [] Notebook prose and register cleanup (markdown structure, final top-to-bottom run).
- [] Report: Introduction and related work (CIFAKE paper, Hendrycks and Dietterich, one or two detector papers).
- [] Report: Methodology and Experimentation sections from the frozen protocol and the CSVs.
- [] Report: Analysis and conclusion (worth 10 of the 30 points, the biggest single prize).
- [] Report: Abstract, references, IEEE formatting pass (8 to 10 pages).
- [] Presentation: skeleton, slides, then the 10-minute recording with both of us on it.